

2.2 Investments involving a single deposit

If \$C\$ is an initial investment for time length T under an effective annual rate i compounded annually, we have

$$A_c(\tau) = C(1+i)^T$$

This gives 4 unknown variables

$A_c(\tau)$ C T i

If any 3 are known we can solve for the remaining variable. Depending on what it is, solving looks different.

(1.) Unknown C

Problem: Aiko's aunt opens an account for her niece. The account earns compound interest at an annual effective interest rate of 8%. The aunt notifies Aiko of the gift but not of the amount deposited. Five years after the account was opened, Aiko closes the account and receives \$3,673.32. What was the amount C that Aiko's aunt deposited?

Solution.

$$i = 8\% , \quad T = 5 , \quad A_C(T) = 3,673.32$$

$$3673.32 = C(1.08)^5 \Rightarrow C = 2500$$

(2.) Unknown Time length

Problem: Chen opens an account with \$1,000. The account earns interest at an annual effective rate of 4%. Chen learns that the balance is currently \$1,342. How long did it take to earn the \$342 of interest?

Solution.

$$C = 1000 \quad i = .04 \quad A_C(T) = 1342$$

$$1342 = 1000(1.04)^T \Rightarrow 1.342 = 1.04^T$$
$$\ln(1.342) = T \ln(1.04)$$

$$T = \frac{\ln(1.342)}{\ln(1.04)} = 7.5 \text{ years}$$

(3) Unknown Rate

Problem: Adrianna deposits \$400 in an account for her daughter that grows by compound interest. After five years the balance has grown to \$463.71. Assume there is a constant annual effective interest rate i earned by the account, and find that rate.

Solution.

$$C = 400, \quad T = 5, \quad A_T(C) = 463.71$$

$$463.71 = 400(1+i)^5$$

$$i = \sqrt[5]{\frac{463.71}{400}} - 1 = 3\%$$

2.3 Investments involving Multiple deposits

More generally an interest theory problem will involve a sequence of contributions

$$\{C_{t_k}\} = C_{t_1}, C_{t_2}, \dots, C_{t_k}, \dots$$

Some contributions can be

(+) deposit

(-) withdrawal

let T be final time (when everything is liquidated)

let B be the balance at time T

let $a(t)$ be the amount function governing the account value.

definition The time τ equation of value

of C_{t_k} contributed at time t_k is

$$C_{t_k} \cdot \frac{a(\tau)}{a(t_k)}$$

← bring forward to time τ

← bring back to time 0

and the time τ value of B is

$$B \cdot \frac{a(\tau)}{a(\tau)}$$

Thus,

$$\sum_{\text{all } k} C_{t_k} \frac{a(\tau)}{a(t_k)} = B \frac{a(\tau)}{a(\tau)}$$

time 0 equation: $\sum_{\text{all } k} C_{t_k} \frac{1}{a(t_k)} = B \cdot \frac{1}{a(\tau)}$

$$= \sum_{\text{all } k} C_{t_k} v(t_k) = B \cdot v(\tau)$$

all k

time T equation: $\sum_{\text{all } k} C_{t_k} \frac{a(T)}{a(t_k)} = B \frac{a(T)}{a(T)}$
 $= B$

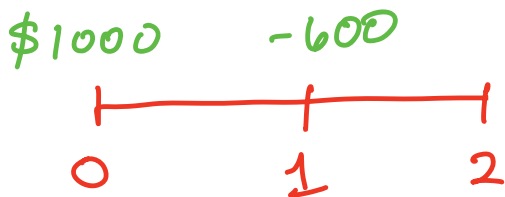
Key

DRAW

TIMELINES!

Problem: John borrows \$1,000. The loan is governed by compound interest at an annual effective interest rate of 10%. John repays \$600 at the end of one year and plans to complete repayment of the loan with a payment of P at the end of the second year. Find P .

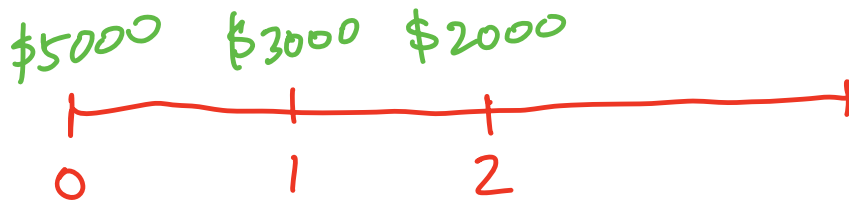
Solution



$$-600(1.1) + 1000(1.1)^2 = 550$$

Problem: Caitlin opens a savings account with a deposit of \$5000. She deposits \$3000 a year later and \$2,000 a year after that. The account grows by compound interest at a constant annual effective rate i . Just after Caitlin's \$2,000 deposit, her balance is \$11,000. Find the effective rate of interest i .

Solution.



at time 2, $B = 11,000$

$$11,000 = 5000(1+i)^2 + 3000(1+i) + 2000$$

$$11 = 5(1+2i+i^2) + 3 + 3i + 2$$

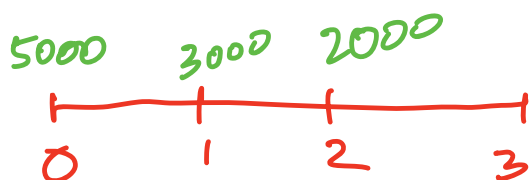
$$0 = 5i^2 + 13i - 1$$

$$i = \frac{-13 \pm \sqrt{13^2 - 4(5)(-1)}}{2(5)}$$

$$= .074772708$$

Problem: Daphne opens a savings account with a deposit of \$5,000. She deposits \$3,000 a year later and \$2,000 a year after that. The account grows by compound interest at a constant annual effective rate i . Three years after Daphne opened the account, it has a balance of \$11,000. Find the effective rate of interest i .

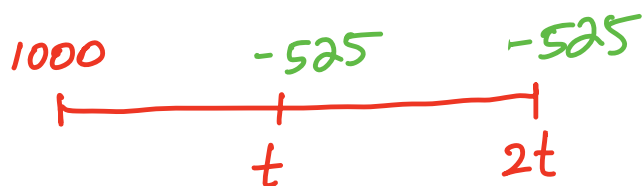
solution.



$$5000(1+i)^3 + 3000(1+i)^2 + 2000(1+i) = 11,000$$

Need a calculator to solve cubic!

Problem: Franklin borrows \$1,000 and the loan is governed by compound interest at an annual effective interest rate of 10%. Franklin agrees to repay the loan by two equally spaced payments of \$525. When should he make these payments?



$$1000 = \frac{525}{(1.1)^t} + \frac{525}{(1.1)^{2t}} = \frac{525}{1.1^t} + \frac{525}{((1.1)^t)^2}$$

$$1000(1.1^t)^2 = 525(1.1^t) + 525$$

$$\text{let } 1.1^t = x$$

$$1000x^2 - 525x - 525 = 0$$

$$x = \frac{525 \pm \sqrt{(-525)^2 - 4(1000)(-525)}}{2000}$$

$$= 1.033153132 \Rightarrow t = \frac{\ln(1.033153132)}{\ln(1.1)}$$

$$t = .342202893$$

$$2t = .684405785$$

2.4 Investment Return

there could be one, many, or no interest rates which satisfy

$$\sum_k C_{t_k} (1+i)^{\tau-t_k} = B(1+i)^{\tau-T}$$

we call such interest rates

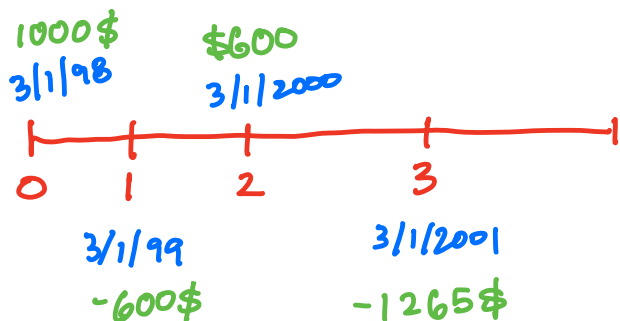
Yield Rates or Internal Rates of Return

↳ measures how attractive a transaction is.

Lender — wants higher yield rates
Borrower — wants lower yield rates

Problem: Gautam invested \$1,000 on March 1, 1998 and \$600 on March 1, 2000. In return he received \$600 on March 1, 1999 and \$1,265 on March 1, 2001. Show that $i = 10\%$ is the unique yield rate.

Solution.



$$1000 + 600(1+i)^{-2} = 600(1+i)^{-1} + 1265(1+i)^{-3} \quad \leftarrow \text{bring to 0}$$

or

$$1000(1+i)^3 + 600(1+i) = 600(1+i)^2 + 1265 \quad \leftarrow \text{bring to } T_{(3)}$$

Let $1+i = x$ and

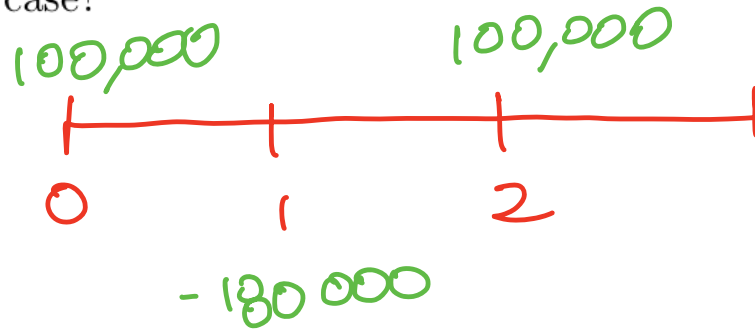
$$p(x) = 1000x^3 - 600x^2 + 600x - 1265 \quad \rightarrow p(1.1) = 0 \quad \checkmark$$

$$\begin{array}{r|rrrr} 1.1 & 1000 & -600 & 600 & -1265 \\ & \downarrow & 1100 & 550 & 1265 \\ \hline & 1000 & 500 & 1150 & 0 \end{array}$$

$$p(x) = (x - 1.1) \underbrace{(1000x^2 + 500x + 1150)}_{\text{Irreducible!!!}}$$

Problem: Ace Manufacturing agrees to pay \$100,000 immediately and again in exactly two years in return for a loan of \$180,000 one year from now (to be used to replace a piece of machinery). Their CEO is asked what yield rate is associated with this transaction, but he is unable to answer the question. Why must this be the case?

Solution.



$$100,000(1+i)^2 - 180,000(1+i) + 100,000 = 0$$

divide by 100,000, let $x = 1+i$

$$p(x) = x^2 - 1.8x + 1$$

$$x = \frac{1.8 \pm \sqrt{1.8^2 - 4(1)(1)}}{2}$$

negative!

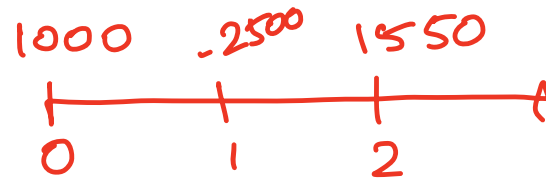
Problem: Banker Johnson is always on the lookout for opportunities to make money without risking any of his own funds. Johnson is able to borrow \$10,000 for one year at an annual effective interest rate of 4%, then loan out the \$10,000 for one year at an annual effective rate of 6%. What is Johnson's yield rate on this transaction?

Solution.

$$\left. \begin{array}{l} 10,000(1.04) = 10,400 \text{ pays} \\ 10,000(1.06) = 10,600 \text{ receives} \end{array} \right\} \text{undefined}$$

Problem: Alice and Afshan are friends. Afshan agrees to give Alice \$1,000 today and \$1,550 two years from now if Alice will give her \$2,500 in one year. What is the yield rate for this transaction?

Solution.



$$1000(1+i)^2 - 2500(1+i) + 1550 = 0$$

$$p(x) = 10x^2 - 25x + 15.5 = 0$$

$$x = \frac{25 \pm \sqrt{25^2 - 4(10)(15.5)}}{20} = 1.361803399$$

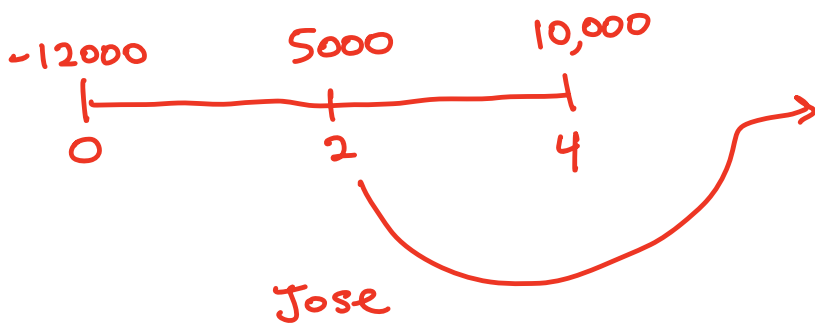
or

$$1.138196601$$

2.5 Reinvestment Considerations

We open this section with an example

Problem: Jose loans Martin \$12,000. Martin repays the loan by paying \$5,000 at the end of two years and \$10,000 at the end of four years. The money received at $t = 2$ is immediately reinvested at an annual effective interest rate of 2.4%. Find Martin's rate of interest and Jose's annual yield rate.



Martin's rate of interest

$$10,000 + 5,000(1+i)^2 = 12,000(1+i)^4 \quad \text{let } (1+i)^2 = x$$

$$10 + 5x = 12x^2 \Rightarrow 12x^2 - 5x - 10 = 0$$

$$x = \frac{5 \pm \sqrt{25 - 4(12)(-10)}}{24} = 1.144675211$$
$$\Rightarrow i = .069894953$$

Jose's Annual yield rate: Bottom line approach

$$t=4 \quad \text{Jose has} \quad 10,000 + 5000(1.024)^2 \\ = 15242.88$$

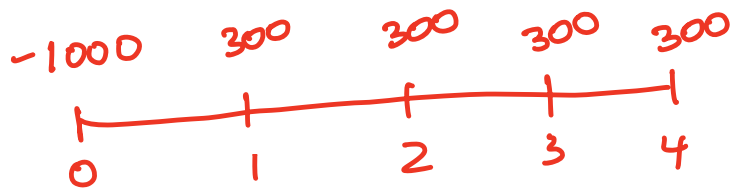
$$\text{So} \quad 12000(1+i_J)^4 = 15242.88$$

$$i_J = .061625755$$

Notice that Jose's yield rate is lower than Martin's since Jose reinvested at an interest rate lower than Martin.

Problem: Julie pays Chan \$1,000. In return, he pays her \$300 at the end of 1, 2, 3, and 4 years. Each time Julie receives a payment, she reinvests it at 3%. She closes the 3% account at the end of 4 years. Find the annual effective interest rate paid by Chan and the annual yield rate earned by Julie.

Solution.



$$1000 = 300 \left((1+i)^{-1} + (1+i)^{-2} + (1+i)^{-3} + (1+i)^{-4} \right) \quad \text{Julie}$$

$$\frac{10}{3} = (x + x^2 + x^3 + x^4) \quad \text{where } x = (1+i)^{-1}$$

by computer/calculator $x = .92839 \Rightarrow$
 $i_C = .077133532$

to find i_J

$$1000 (1+i_J)^4 = 300(1.03)^3 + 300(1.03)^2 + 300(1.03) + 300$$

$$i_J = .058445627$$